

Syllabus

Fall semester 2025

Course Title	Basic Physics 2		Course Code		Credit		3	Classroom	5510
			Type		Hours	3			
Teacher	Name	Class no.	Office	Office hours	Contact		E-mail		
	Hanyi Yang	1	5505	TUE,WED,THU,FRI 1~2	chickeny@ksa.hs.kr				
	Google Classroom Code: yhc63aje								

1. Course Description

- Areas of study cover Oscillation and Waves, Geometrical Optics and Electricity.
- The main concepts are electric charge, electric field, electric potential, electric current and circuits, magnetic field, electromagnetic induction and light.
- Classes contain regular instructions (theory and problem solving) and demos which are firmly integrated into the course. Each 50 min period of regular classes generally consist of a 15-25 minutes concept review of reading assignment including applications and demonstrations, followed by 20-30 minutes of applied problem solving. The problems may be reviewing ones that were assigned for homework on a previous day, or working in class on ones that are assigned for the following day. A systematic approach to problem solving is modeled and encouraged throughout the year. Students are generally allowed to work in small groups (teams) to help each other, as well as getting individual help from the instructor. Students are allowed to cooperate in completing homework and studying course material.
- Assessments in this class are divided into three categories: homework, tests and exams. Homework assess the students' abilities to work with textbook, and use their notes and knowledge they learned in class to solve problems. Tests and exams assess the abilities to understand and apply concepts to problem solving.
- The homework will be assessed by means of submission of solutions the assigned problems.
- The tests will consist of free response and multiple choice questions and model Midterm and Final Exams.

2. Core Target Competencies

Subject Core Competencies			
Critical Thinking Skills	Analytical and Creative Thinking	Complex Problem-Solving Skills	Creativity
Fundamental concepts in several subjects in physics will be presented, including electromagnetism, optics, and subjects in modern physics, are introduced. Following the homework and the discussion sessions in this lecture, students will be encouraged to develop their competencies listed above. The ways of thinking, trained in this lecture, will provide important background for future coursework.			

3. Textbooks and References

Textbook: Physics by Giancoli, 6th edition.

4. Class Method

- A. Self-learning during homework, assessment through short quizzes.
- B. The lectures will provide students with various approaches to the subject that differ from the textbook.
- C. Wide use of physical demonstrations and computer simulations.

5. Evaluation

A. Evaluation Criteria

Midterm	Final	Quiz	Homework	Attendance and Attitude	Total
30	30	20	10	10	100

B. Evaluation Method

- (1) The midterm and final exams are given through written tests. The problems consist of short answer questions and free response questions. Problems are designed to understanding the logical structure of the subject, and solving the problems using the acquired knowledge.
- (2) Quizzes will be given two times during the semester. The problems consist of multiple choice questions and short answer questions. Problems are designed to emphasize memorization of the basic concepts.
- (3) Homework: Penalty will be given to the overdue homework.
- (4) 1 point penalty per absence without notice. 3 tardies are treated as 1 absence.

6. Criteria for Awarding Credits

👉 Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal’s Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

삭제됨:

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- Calculation of the basis score for awarding credit to absentees : Convert the Z-score of the test to the score of the non-test.
- If a basis score cannot be calculated, the grading method : Follow the grading regulations.
- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week	Topic and Contents	Homework
Week 1	Orientation	
Week 2	Ch. 20 Static Electricity (Electric Charge, Electric Force, Electrostatic induction)	36, 48, 51, 57, 61
Week 3	Ch. 21-1 Creating and Measuring Electric Fields (Electric Fields)	58, 67, 71, 73
Week 4	Ch. 21-2 Applications of Electric Fields (Electric potential, <u>Capacitor</u>)	76, 81, 86, 92, 95
Week 5	Ch. 22 Current Electricity (Electric energy, Power, Voltage, Resistor)	64, 67, 71, 77, 79
Week 6	Ch. 23 Series and Parallel Circuits	65, 67, 77, 81, 82
Week 7	Review & Quiz	National Foundation Day (10.03.)
Week 8	Chuseok Holiday, Hangeul Day, Discretionary Holiday	
Week 9	Midterm Exam	
Week 10	Ch. 24-1 Magnets: Permanent and Temporary (Electric current, Magnetic Field, Magnetics)	51, 58, 64, 66
Week 11	Ch. 24-2 Force Caused by Magnetic Fields (Magnetic Force, <u>Lorentz Force</u>)	72, 78, 88, 90, 95
Week 12	Ch. 25-1 Electric Current from Changing Magnetic Fields (<u>Faraday's Law</u> , Lenz's Law)	49, 62, 68, 70, 73,
Week 13	Ch. 25-2 Changing Magnetic Fields Induce EMF (<u>Transformer</u>)	50, 80, 87, 91
Week 14	Ch. 14-2 Wave Properties	53, 68, 76, 83, 87
Week 15	Ch. 15 Sound Review & Quiz	51, 56, 66, 75, 80
Week 16		Make-up class week
Week 17	Final Exam	

※ The above plans may change according to the academic schedule.